GlandersWatch: A Chained Multi-Level Machine Learning Framework for Population-Level Glanders Risk Prediction

Project Report Submitted

in partial fulfillment of the requirements for the award of the

Bachelor of Technology

in

Computer Engineering

by

Yuvraj Singh XXXXXXXXX

Under the supervision of

Dr. [Supervisor Name]



Department of Computer Science and Engineering

International Institute of Information Technology, Bhubaneswar

November 2026

Certificate

This is to certify that the report entitled "GlandersWatch: A Chained Multi-Level Machine Learning Framework for Population-Level Glanders Risk Prediction" submitted by Yuvraj Singh (XXXXXXXXX) to International Institute of Information Technology, Bhubaneswar is a record of bonafide project work under my supervision, and the report is submitted for the evaluation of the BTech Degree Final-Semester Project.

(Supervisor)
Dr. [Supervisor Name]

Declaration

I certify that

1. The work contained in the report has been done by myself under the general supervision of my supervisor.
2. The work has not been submitted to any other Institute for any degree or diploma.
3. I have followed the guidelines provided by the Institute in writing the report.
4. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
5. Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references.
6. Whenever I have quoted written materials from other sources, I have put them under quotation marks and given due credit to the sources by citing them and giving required details in the references.

Yuvraj Singh
XXXXXXXXX

Acknowledgments

I would like to express my sincere gratitude to my respected supervisor, Dr. [Supervisor Name], for the invaluable guidance, support and encouragement given throughout this project. Their insights and expertise were instrumental in shaping this work and helped me overcome numerous challenges. I am also thankful to the faculty and staff of the Department of Computer Science and Engineering, International Institute of Information Technology, Bhubaneswar, for providing the resources and a conducive learning environment. Finally, I would like to thank my family and friends for their unwavering support and understanding during the course of this project.

Yuvraj Singh
XXXXXXXXX

Abstract

Glanders is a bacterial zoonosis of equids caused by *Burkholderia mallei*. It is notifiable, frequently fatal, and remains endemic in parts of Africa, Asia and South America. Indian surveillance between 2019 and 2023 identified 728 seropositive equids across 138 districts, with strong geographic concentration. The published machine learning literature on this disease is confined to laboratory identification of the organism by spectroscopic methods. No population-level geographic risk model has been reported.

This project develops such a model. Four public datasets that had not previously been applied to glanders are combined into a country-year panel: a text-mined occurrence corpus from Europe PMC, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology. The panel contains 3,078 country-year observations across 162 equine-holding countries for the years 2005 to 2023, with 172 positive labels and 44 engineered features.

Six model families are trained under three validation schemes, and a three-stage chained pipeline is evaluated in which animal population is forecast before occurrence is predicted. On a forward temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 0.056, an approximately ten-fold lift, with an expected calibration error of 0.016. Performance falls to 0.304 under leave-country-out validation, which is the honest estimate for an unseen country. Chaining produces no measurable gain in discrimination. An ablation shows that climate covariates transfer poorly across countries, and the model performs least well in low-income regions. The dominant limitation is that the label reflects documented activity rather than true incidence, so the model largely learns which countries report disease. Reporting effort is identified as the principal confounder for future work.

Keywords: Glanders, *Burkholderia mallei*, Zoonosis, Risk Prediction, Class Imbalance, Gradient Boosting, Probability Calibration, SHAP, Text Mining, Animal Health Surveillance.

Contents

Abstract **4**

1 Introduction **7**

1. Background . . . 7
2. Problem Definition . . . 7
3. Objectives . . . 8
4. Scope and Limitations . . . 8
5. Organisation of the Report . . . 8

2 Literature Review and Gap Analysis **9**

1. Glanders Epidemiology and Burden . . . 9
2. Surveillance in India . . . 9
3. Geographic and Spatiotemporal Risk Analysis . . . 9
4. Machine Learning for Animal Disease Risk . . . 10
5. Machine Learning and Glanders . . . 10
6. Text Mining for Disease Event Extraction . . . 10
7. Prior Art Audit . . . 10
8. Research Gap . . . 10

3 Methodology **12**

1. Data Sources . . . 12
2. Panel Construction . . . 13
3. Preprocessing . . . 13
   - 3.1 Duplicate Removal . . . 13
   - 3.2 Missing Values . . . 13
   - 3.3 Treatment of Absence . . . 13
4. Label Definition . . . 13
5. Feature Engineering . . . 14
6. Feature Selection . . . 14
7. Models . . . 14
8. Chained Pipeline . . . 15
9. Evaluation Design . . . 15
10. Leakage Controls . . . 16

4 Implementation **17**

1. Software Architecture . . . 17
2. Training Pipeline . . . 17
3. Evaluation Pipeline . . . 17
4. Alert Engine Design . . . 17

5. Reproducibility 19

5 Results and Discussion **20**

1. Dataset Characteristics 20
2. Statistical Association and Feature Selection 21
3. Model Comparison 23
4. Calibration 24
5. Temporal Resolution 24
6. Ablation 24
7. Chained Pipeline 26
8. Alert Economics 26
9. Subgroup Analysis 27
10. Discussion 27

6 Conclusion and Future Work **28**

1. Conclusion 28
2. Future Work 28

Bibliography **30**

Chapter 1

Introduction

1. Background

Glanders is a contagious and potentially fatal disease of horses, donkeys and mules caused by the Gram-negative bacterium *Burkholderia mallei*. The organism has no environmental reservoir. Infection persists through transmission among equids, and asymptomatic carriers can remain infectious for extended periods. Human infection is rare but carries a high case fatality rate, and occurs through occupational contact with infected animals. The disease is notifiable to the World Organisation for Animal Health and is classified as a potential bioterrorism agent.

The economic significance of glanders is concentrated in low and middle income countries, where equids remain essential to transport, construction, agriculture and brick manufacture. India recorded 5.44 lakh equids in the 2019 Livestock Census, a large proportion held by households that depend on the animals for daily income. For such households the loss of a working animal is a direct loss of livelihood, which makes early warning valuable even where mass treatment is not feasible.

Surveillance data indicate that the disease is neither rare nor geographically contained. A national programme screening 158,798 equid serum samples from 391 districts between 2019 and 2023 identified 728 seropositive animals in 138 districts, with roughly sixty per cent of positives reported from Uttar Pradesh. Historical surveillance covering 2006 to 2018 reported 1,145 positive equids among 128,404 sera tested. These figures describe a persistent, spatially structured problem rather than isolated outbreaks.

2. Problem Definition

The operational question facing a district veterinary officer is not whether glanders exists in the country, but whether it is likely to appear in a given area in the coming year. Answering it requires combining several kinds of evidence: the history of the disease in and around the area, the size of the susceptible population, and the environmental conditions that permit transmission.

Surveillance systems report what has already happened. Forecasting systems exist for several livestock diseases but not for glanders. The problem addressed here is therefore the construction and honest evaluation of a population-level risk model for glanders using only publicly accessible data.

The problem has four parts.

1. **Data assembly.** No single public source records glanders occurrence completely. Occurrence must be reconstructed from the literature and joined to population and environmental covariates at a common geographic and temporal resolution.

2. **Label construction.** The outcome is rare, which makes naive accuracy useless and requires an explicit treatment of what a negative observation means.
3. **Model selection under distribution shift.** A model fitted on some countries must be evaluated on countries it has never seen, because that is the deployment condition.
4. **Threshold economics.** A probability is not an alert. The operating point must be chosen so that the volume of warnings is defensible relative to the events caught.

3. Objectives

The objectives of the project are:

1. To assemble a country-year panel for glanders from public sources not previously applied to this disease, covering animal history, animal population and environment.
2. To engineer and select features using statistical evidence rather than assumption, and to document which evidence was used and why.
3. To train and compare six model families under three validation schemes, with the harder scheme governing model selection.
4. To design and evaluate a chained pipeline in which animal population is forecast as an intermediate step.
5. To produce calibrated probabilities and an alert operating curve, so that the output can support a warning decision rather than only a ranking.
6. To report limitations and negative results explicitly, including any component that failed to improve performance.

4. Scope and Limitations

The study is scoped to country-year resolution. District-level modelling, which the original project proposal envisaged for India, is not attempted here because the required district-level covariates could not be assembled from public sources within the project period.

The model is a population-level risk tool. It does not diagnose an individual animal and is not a substitute for laboratory confirmation or clinical examination.

Three limitations are stated at the outset because they shape every result that follows. First, the label is derived from published reports, so a negative observation may indicate absence of disease or absence of reporting. Second, occurrence is reconstructed by an automated extractor that has not been validated against a hand-annotated reference set. Third, with 172 positive observations the statistical uncertainty on every headline metric is substantial, and this is quantified rather than hidden.

5. Organisation of the Report

Chapter 2 reviews the epidemiological and machine learning literature and identifies the gap addressed here. Chapter 3 describes the data, the panel construction, the feature engineering and the modelling approach. Chapter 4 covers the implementation, including the software architecture and both computational pipelines. Chapter 5 presents and discusses the results. Chapter 6 concludes and sets out future work.

Chapter 2

Literature Review and Gap Analysis

1. Glanders Epidemiology and Burden

Glanders is reviewed in the veterinary literature as a neglected zoonosis with high case fatality and a genuine occupational risk to handlers [1, 2]. A recent systematic review and meta-analysis consolidates seroprevalence estimates across published studies and identifies management practice and animal movement as the dominant epidemiological drivers [3]. Regional reviews document continued circulation in Brazil [4], and the disease remains endemic or repeatedly reported across parts of Africa, Asia and the Middle East.

The consistent finding across this literature is that glanders persists where equid populations are large, where animals move frequently, and where surveillance is weak. The first two conditions are measurable from public data. The third affects whether the disease is visible at all, and is therefore a confounder rather than a covariate.

2. Surveillance in India

Indian surveillance provides the strongest evidence base for the disease. Malik et al. described the emergence and re-emergence of glanders in India through outbreaks reported between 2006 and 2011 [7]. Singha et al. reported serological surveillance and clinical investigation of indigenous equines for the period 2015 to 2018 [6]. More recently, the national programme covering 2019 to 2023 screened samples from 391 districts and produced a prevalence estimate together with a risk analysis [5].

These studies establish where glanders occurs, which districts carry the burden, and which animal-level factors are associated with seropositivity. None constructs a predictive model, and none uses population or reanalysis covariates as predictors.

3. Geographic and Spatiotemporal Risk Analysis

Descriptive spatial analysis of glanders has been reported. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian glanders data [8]. More broadly, species distribution and ecological niche models have been compared as tools for livestock and disease epidemics [9].

This body of work is descriptive. Cluster detection identifies where cases have already concentrated. It does not forecast which cells will be positive in a future period, and that distinction matters for an early warning application, where the value lies in anticipating rather than confirming.

4. Machine Learning for Animal Disease Risk

Predictive modelling of animal disease risk is established for avian influenza. Kane et al. compared ARIMA and random forest models for the prediction of H5N1 outbreaks [11]. Walsh et al. used machine learning to target active surveillance for avian influenza viruses [12]. Yoo et al. developed dynamic farm-level risk prediction for highly pathogenic avian influenza in the Republic of Korea [13], and Alshatti et al. produced a risk prediction model for outbreaks in Kuwait [10].

This literature supplies the methodological template used here: assemble occurrence, population and environmental covariates at a common spatial unit, fit tree-based ensembles, and evaluate on held-out periods. The template has been applied to influenza, and not to glanders.

5. Machine Learning and Glanders

The only machine learning work specific to *Burkholderia mallei* that could be located is spectroscopic identification of the organism. Moawad et al. developed a Raman spectroscopic assay in which learned models classify spectra to identify *B. mallei* and related species [14]. Stöckel et al. reported Raman spectroscopic detection and identification of *B. mallei* and *B. pseudomallei* [15].

Both are laboratory diagnostic problems. Neither addresses the geographic or temporal distribution of the disease, and neither produces a risk estimate for a population.

6. Text Mining for Disease Event Extraction

Using published literature as a structured source of occurrence data has precedent in epidemic intelligence. Collier surveyed web-based epidemic intelligence and the role of text mining within it [16], and Lejeune et al. described multilingual event extraction for epidemic detection [17].

The present work applies sentence-level event extraction not as a supplementary signal but as the primary source of the label, for the pragmatic reason that official outbreak records are incomplete for many countries. The consequence, that the label inherits both the strengths and the reporting biases of the literature, is analysed rather than assumed away.

7. Prior Art Audit

To test the novelty claim rather than assert it, a reproducible set of queries was run against Europe PMC. Each row of Table 2.1 is a query count, and the full query strings and retrieved titles are retained alongside the project code.

The decisive cells are the query pairing glanders with NASA POWER, which returns zero records, and the pairing with gradient boosting, which returns four, none of which is a glanders risk model. The nearest prior work is descriptive cluster detection in Brazil and a risk-factor analysis of the Indian serosurvey.

8. Research Gap

The literature reviewed above divides into three disjoint groups. Epidemiological studies describe where glanders occurs. Machine learning studies of *B. mallei* classify laboratory spectra. Veterinary risk models exist but are built for influenza. No study joins an occurrence corpus, a livestock census, development

Table 2.1: Europe PMC hit counts for combinations of glanders with each data source and method used in this study.

Query	Hits
glanders (baseline corpus)	6674
glanders + machine learning	75
glanders + tree ensembles	40
glanders + forecasting	139
glanders + FAOSTAT	7
glanders + NASA POWER	0
glanders + World Bank indicators	22
glanders + equine population denominator	54
glanders + Europe PMC text mining	378
glanders + chained/stacked models	36
glanders + random forest (risk context)	8
glanders + niche / distribution model	62
glanders + remote sensing / reanalysis	10
glanders + spatio-temporal clustering	463
glanders + gradient boosting	4

indicators and reanalysis climatology into a single panel for glanders, and none evaluates a calibrated predictive model on that panel.

That gap defines the contribution of this project.

Chapter 3

Methodology

1. Data Sources

Four publicly accessible sources are used. None had previously been applied to glanders risk modelling, and each contributes a distinct kind of evidence.

Table 3.1: Data sources and their role in the panel.

Source	Role	Content	Coverage
Europe PMC	Animal history	Occurrence events extracted from abstracts	6,674 records, 263 events
FAOSTAT	Animal population	Stocks of horses, asses, mules, camels, cattle, sheep and goats	68,449 records, 1961–2023
World Bank	Socio-economic context	Ten development indicators and country centroids	217 countries, 2000–2024
NASA POWER	Environment	Temperature, precipitation, humidity and wind	Six measures, 211 centroids, 2005–2024

Figure 3.1 shows the distribution of the extracted occurrence events by year and by country. The volume of recorded activity rises after 2015, which reflects growth in open access veterinary publishing as much as any change in disease incidence. This is one reason the label is treated as documented activity throughout.

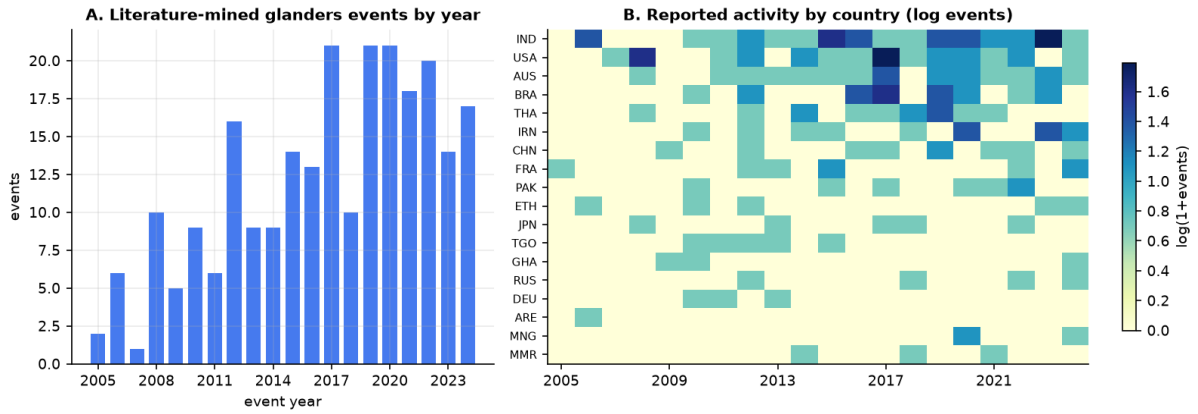


Figure 3.1: Occurrence events extracted from Europe PMC. Panel A gives the annual count. Panel B gives the distribution across the most frequently reporting countries on a logarithmic scale.

2. Panel Construction

All four sources are keyed on ISO 3166 alpha-3 country code and calendar year, so they join without ambiguity once names are reconciled. FAOSTAT is streamed from the 32 MB compressed bulk archive, which expands to approximately 545 MB, and filtered to the seven stock series of interest rather than loaded into memory.

Country names are matched to ISO-3 through `pycountry` with explicit overrides for non-standard forms such as "Türkiye", "Viet Nam" and "Côte d'Ivoire". World Bank regional and global aggregates are excluded by requiring a three-character ISO code, which prevents rows such as "World" or "South Asia" from entering the panel as if they were countries.

The full grid contains 6,700 country-year rows across 268 countries for 2000 to 2024. For modelling, three restrictions are applied: only equine-holding countries are retained, defined as a maximum equid stock of at least one thousand animals; only the years 2005 to 2023 are used; and only rows with an observable next-year label are kept. The resulting matrix has 3,078 rows, 44 features and 172 positive labels.

3. Preprocessing

3.1 Duplicate Removal

Extracted events are de-duplicated on the composite key of country, year and source record identifier. Without this step a single paper reporting several cases would contribute multiple counts and inflate the occurrence signal.

3.2 Missing Values

Missing numeric values are imputed with the column median, and every column that required imputation keeps a binary missing-indicator feature so that absence is represented rather than silently filled. Imputation is fitted inside the training pipeline, so the test period cannot influence imputed values. Missingness is low: 0.00% in the FAOSTAT block, 1.68% in the World Bank block, 2.19% in the climate block, and 0.99% overall, with no column exceeding 5.8%.

3.3 Treatment of Absence

A country-year is assigned a negative label only after the country passes the equine-stock filter. This prevents a country with no recorded equids from contributing an uninformative negative observation, and ensures that a blank is never read as proof that disease was absent.

4. Label Definition

The prediction target is binary: whether a glanders occurrence was documented in country i during year $t+1$. Occurrence is extracted from Europe PMC by matching, within a single sentence, a country name from a curated gazetteer and a term from an occurrence lexicon containing *outbreak*, *case*, *detected*, *seropositive* and *surveillance*. When both are present, an event is recorded for the year stated in the sentence, or the publication year where no year is stated.

Because the label depends on publication it is termed *documented activity* throughout. A negative value may indicate either the absence of disease or the absence of a report, and the model is therefore interpreted as a predictor of surveillance-observable activity.

5. Feature Engineering

Four groups of features are constructed, all using information available at time t or earlier.

Table 3.2: Feature groups, sizes and rationale.

Group	Count	Content and rationale
Disease history	7	Occurrence in the two prior years, a five-year running total, years since the last event, and contemporaneous world and regional activity shares. The disease persists in an area once introduced.
Animal population	14	Log equid and total livestock stocks, equid share of farm animals, equid density, one- and three-year growth, and the seven species counts. A larger susceptible population offers more opportunity for detection.
Climate	10	Mean, maximum and minimum temperature, precipitation total and coefficient of variation, relative humidity and wind, each paired with an anomaly computed against the country's own expanding mean. The anomaly separates unusual conditions from latitude.
Socio-geography	11	Income band, agricultural land share, agricultural value added, livestock production index, GDP per capita, rural share, internet penetration, fertiliser intensity, life expectancy, population and density. These largely proxy surveillance capacity.

Three derived features require definition. Equid share is the sum of horses, asses and mules divided by total farm animals. Livestock units apply metabolic weights, with cattle and camels at 0.8 and small ruminants at 0.1, so that one number expresses grazing pressure across species. Years since the last event uses the value 99 to denote no recorded event within the panel window.

6. Feature Selection

Selection proceeded in two stages. First, columns with no measurable association with the label were removed. Mutual information and Spearman correlation were computed for every numeric feature, and columns with mutual information below 0.002 and absolute correlation below 0.02 were discarded. Second, redundancy was addressed. One member of each pair with absolute correlation at or above 0.90 was removed, and a column that was exactly the sum of three others was dropped because a tree model can construct that sum itself.

No linear projection or principal component analysis was applied. With 44 features and 3,078 rows the binding constraint proved to be signal rather than dimensionality. This is supported by the ablation in Section 6., which shows that reduced feature sets generalise better across countries than the full set.

The selection criterion is itself examined in Section 2., where marginal and conditional importance are compared.

7. Models

Six model families are compared. The set spans the range from additive to fully interactive, and includes a deliberately simple reference.

Logistic regression provides an additive baseline with balanced class weights. A single decision tree is included because shallow trees are interpretable by a non-specialist, and because the performance gap between one tree and an ensemble is itself informative. Random forest and extra trees are bagged ensembles, differing in the degree of randomisation applied to split selection. XGBoost and LightGBM are boosted ensembles, differing in tree growth strategy [18, 19].

Table 3.3: Model families and the hyperparameters selected by tuning.

Model	Family	Selected configuration
Logistic regression	linear	Additive model with balanced class weights; the reference baseline.
		Tuned settings: $C = 1.0$, ℓ_2 penalty, balanced class weights
Decision tree	tree	Single shallow tree; interpretable, and a low-variance check on ensemble gains.
		Tuned settings: depth 3, minimum leaf size 1, criterion entropy
Random forest	bagged trees	Bagged trees over random subsets; variance control at small positive counts.
		Tuned settings: depth 6, minimum leaf size 1, features per split square root of the column count
Extra trees	bagged trees	Bagged trees with randomised split thresholds; further decorrelation when positives are scarce.
		Tuned settings: depth 6, minimum leaf size 10, features per split 0.3
XGBoost	boosted trees	Boosted trees with explicit regularisation.
		Tuned settings: depth 3, regularisation $\lambda = 1.0$, minimum child weight 5
LightGBM	boosted trees	Leaf-wise boosted trees; efficient comparison on wide tabular data.
		Tuned settings: leaves 15, minimum child samples 40, regularisation $\lambda = 1.0$

Neural networks were excluded on evidential grounds. With 172 positive observations and 44 features, a network has far more parameters to estimate than it has positive examples from which to learn, and would be expected to memorise the training period and fail on new data.

8. Chained Pipeline

The project proposal specified a multi-stage design. Three stages are implemented.

1. **Population forecast.** Next-year log equid stock is predicted from the current year's covariates using ridge regression and histogram gradient boosting, evaluated by grouped cross-validation on country.
2. **Occurrence.** The binary label is modelled with the forecast population appended as an additional feature, and compared against the same model without it.
3. **Intensity.** Conditional on an event, log event count is modelled by a random forest regressor trained on positive observations only.

The stages are combined into a hurdle model, in which expected event burden is the product of event probability and conditional intensity.

9. Evaluation Design

Three validation schemes are reported, each answering a different question.

- **Temporal holdout.** Training on 2005 to 2017 and testing on 2018 to 2023, with the test period never observed during fitting. This answers whether the model can be used in the following years.

- **Leave-country-out.** Five folds grouped by country, so every prediction is made for a country absent from training. This answers whether the model transfers to a new country, and it governs model selection because it is the harsher test and the closer approximation of deployment.
- **Repeated stratified cross-validation.** Five folds repeated three times, used as a stability check on the ranking metrics.

Precision-recall AUC is the primary metric. At 5.59% prevalence, ROC-AUC is dominated by the true-negative mass and overstates performance, a well-characterised property of imbalanced problems [22, 23]. ROC-AUC is reported for comparability. Brier score and expected calibration error are reported because an uncalibrated probability cannot support a threshold-based alert [25, 26]. Precision, recall, specificity, F1 and the Matthews correlation coefficient are computed at a threshold tuned on training data only, and bootstrap percentile intervals with one thousand resamples quantify sampling uncertainty.

Grouped rather than random splitting is essential. A random row split would place the same country in both training and test partitions, allowing the model to memorise country identity. Given the geographic proxy effects described in Section 2., this would inflate all reported scores [24].

10. Leakage Controls

Six controls are applied.

1. All features are built from year t or earlier, and only the target columns use $t+1$.
2. The split is chronological, so no future observation can enter training.
3. Hyperparameters are tuned inside training folds using grouped cross-validation.
4. Probability calibration is fitted on training folds only.
5. The alert threshold is fixed on the training period and applied unchanged to test.
6. Missing-indicator features and imputation values are generated inside the pipeline, so the test period cannot influence them.

Chapter 4

Implementation

1. Software Architecture

The project is implemented as a sequence of numbered scripts, each producing artefacts consumed by the next. All intermediate outputs are cached, so any stage can be re-run without repeating the stages before it.

Table 4.1: Processing stages and their outputs.

Script	Stage	Output
`00_novelty_audit`	Prior art audit	Query counts and retrieved titles
`01_fetch_data`	Ingestion	Europe PMC, World Bank and NASA POWER payloads
`01c_extract_faostat`	Ingestion	Seven FAOSTAT stock series
`02_build_panel`	Assembly	Panel, extracted events and audit block
`03_train_evaluate`	Modelling	Metrics, predictions and model comparison
`05_dataset_analysis`	Analysis	Association, redundancy and selection statistics
`04_report`, `06_analysis_figures`	Reporting	Figures and summary tables
`09_make_full_report`	Reporting	This document

2. Training Pipeline

Figure 4.1 shows the training sequence. The chronological split is placed before any model sees the data, so the test years cannot influence fitting, tuning or calibration.

3. Evaluation Pipeline

Figure 4.2 shows the evaluation sequence. One trained model is scored under three schemes, each producing a different answer, and the resulting metrics are summarised with uncertainty intervals.

4. Alert Engine Design

The wider platform for which this model was built includes an alert service that converts a predicted probability into a notification for registered owners and field workers. The rules engine is shown in Fig-

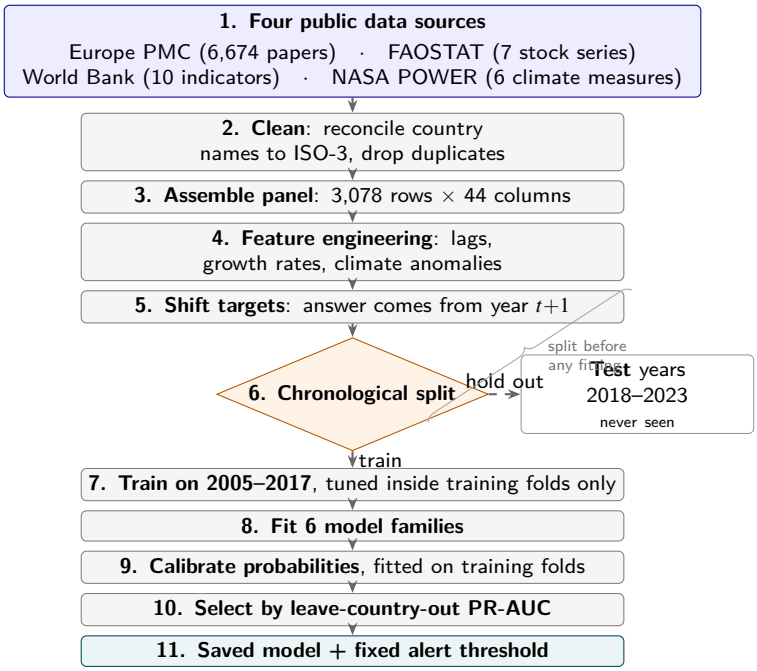


Figure 4.1: Training pipeline. The chronological split sits before any model sees the data, so the 2018–2023 years never influence training. Tuning and calibration both happen strictly inside the training years.

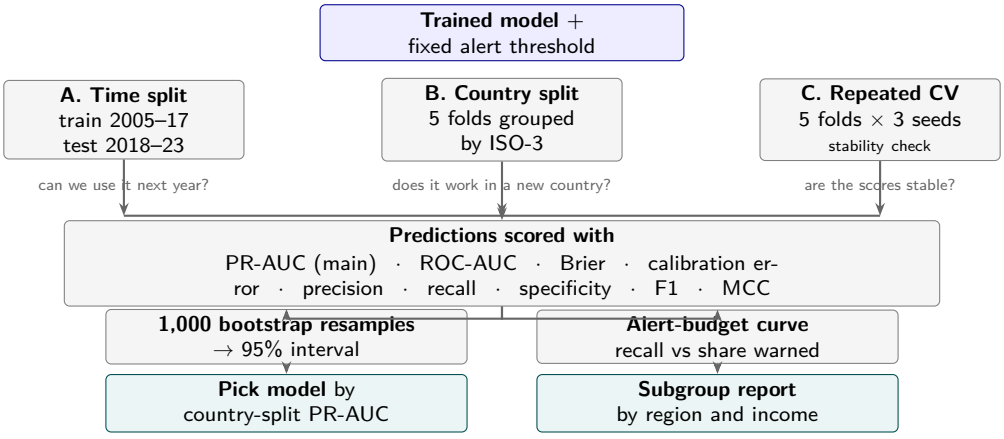


Figure 4.2: Evaluation pipeline. One trained model is scored three different ways, each answering a different question, then reported with uncertainty intervals.

ure 4.3. Three rules govern whether a notification is emitted. A threshold maps the calibrated probability to a risk level. A suppression rule decides whether a message is warranted at all, based on state changes, cooldowns and a data-freshness guard. A deduplication rule prevents the same alert being issued twice for the same area, period and level.

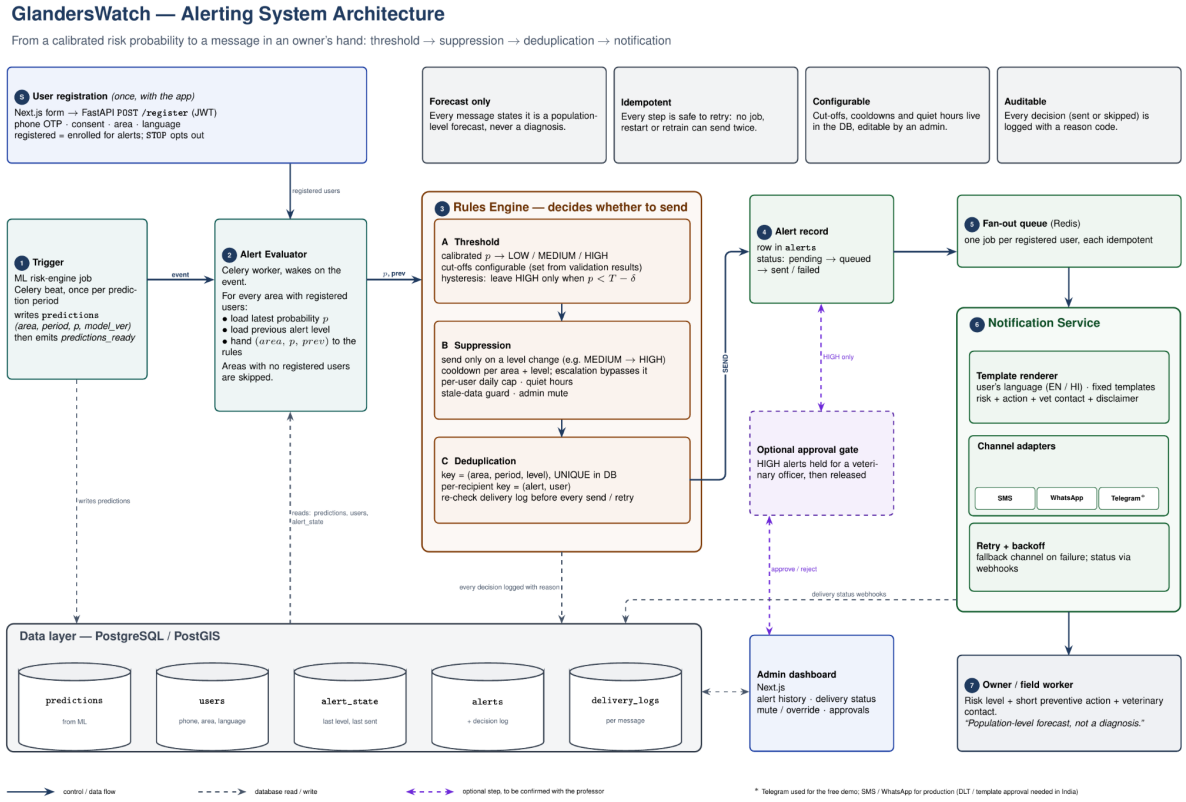


Figure 4.3: Architecture of the alert rules engine, reproduced from the project proposal. The calibrated probability enters the threshold, suppression and deduplication rules before an alert record is created and fanned out to notification channels.

The thresholds used by this engine are taken from the operating curve in Section 8. rather than from a default value. This is the practical link between the modelling work reported here and the alerting behaviour of the platform.

5. Reproducibility

The full computation runs in approximately 4.5 minutes on a laptop processor without a graphics accelerator, using an Apple M4 processor with 10 cores and 16 GB of unified memory. Per-family fitting time at the country-year level ranges from 1.9 seconds for the decision tree to 50.7 seconds for LightGBM.

A single module-level seed constant governs model fitting, cross-validation splitting, bootstrap resampling and SHAP sampling. Software versions are Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1, NumPy 2.5.4, pandas 3.1.1, SciPy 1.18.1. Raw payloads are cached under `data/raw` and the derived panel is written to `data/processed`, so every reported number can be regenerated without re-downloading upstream sources.

One determinism caveat applies. Tree ensembles are seeded but not bitwise reproducible across different thread counts, so parallel execution may change the final decimal of an ensemble score.

Chapter 5

Results and Discussion

1. Dataset Characteristics

The modelling matrix contains 3,078 country-year observations, of which 172 are positive, a prevalence of 5.59% and an imbalance ratio of 16.9 to one. Table 5.1 summarises the distribution.

Table 5.1: Characteristics of the label.

Property	Value
Observations	3,078
Positives and negatives	172 and 2,906
Prevalence	5.59%
Imbalance ratio	16.9 : 1
Countries with at least one positive	56 of 162
Countries positive in two or more years	30
Share of positives in the top five countries	35.5%

Three properties of this distribution matter. The imbalance rules out accuracy as a metric and motivates precision-recall analysis. The temporal persistence of the label, with 30 of 56 positive countries recording positives in more than one year, justifies the use of historical features. The geographic concentration of positives, with the top five countries holding 35.5% of them, provides the first indication that the label tracks reporting capacity, since those countries operate reference laboratories.

Figure 5.1 presents the same information graphically, together with the regional distribution of prevalence.

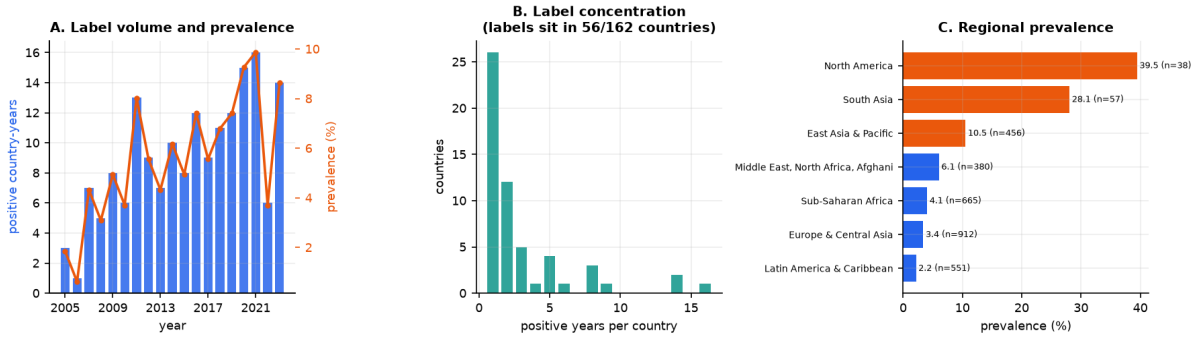


Figure 5.1: Label anatomy. Panel A gives annual positive counts with prevalence on the secondary axis. Panel B gives the distribution of positive years per country. Panel C gives prevalence by World Bank region.

2. Statistical Association and Feature Selection

Spearman correlation, mutual information and Cramér's V were computed for every candidate feature before any model was fitted. The strongest monotone association is `years_since_last` at $\rho = -0.35$, negative because a longer interval since the last recorded event implies lower risk. This is the most defensible epidemiological signal in the panel. Among the categorical strata, region yields a Cramér's V of 0.243, a small association, and income band yields 0.047, which is negligible.

Redundancy is substantial. Variance inflation reaches 228 for `temp_mean`, because annual mean, minimum and maximum temperature are near-duplicates at country resolution. Figure 5.2 shows the correlation structure of the highest-ranked features.

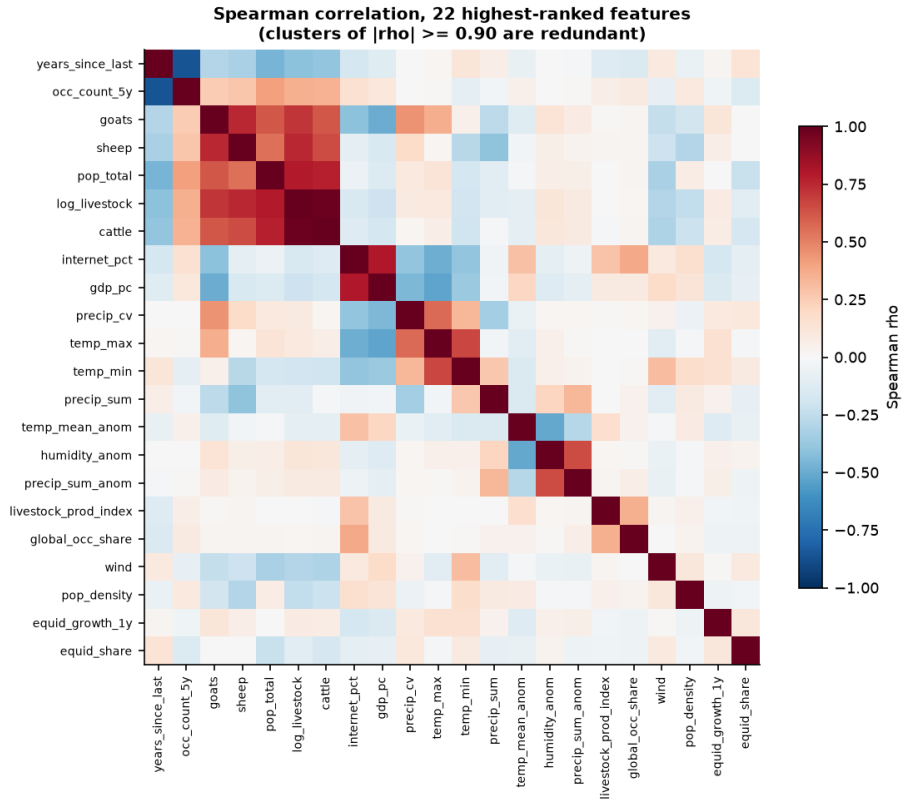


Figure 5.2: Spearman correlation matrix of the twenty-two highest-ranked features, hierarchically ordered. Block structure indicates the collinear groups that motivated redundancy reduction.

The most consequential finding in this section concerns the choice of selection criterion. Marginal association and conditional importance disagree almost completely. The rank correlation between mutual information and SHAP attribution is -0.03, and only 5 of the ten highest-ranked features are shared between the two orderings. `longitude` ranks first on marginal correlation and approximately twentieth on conditional importance, because once population and history are conditioned upon it carries little additional information. It functions as a proxy for which part of the world a country occupies, and therefore for publishing activity.

Figure 5.3 shows the disagreement directly, and Table 5.2 gives the ranked features with both sets of statistics. Ranking features by marginal association alone would favour columns that track reporting practice rather than disease occurrence, so selection was performed on conditional evidence instead.

Figure 5.4 gives the global attribution for the full panel.

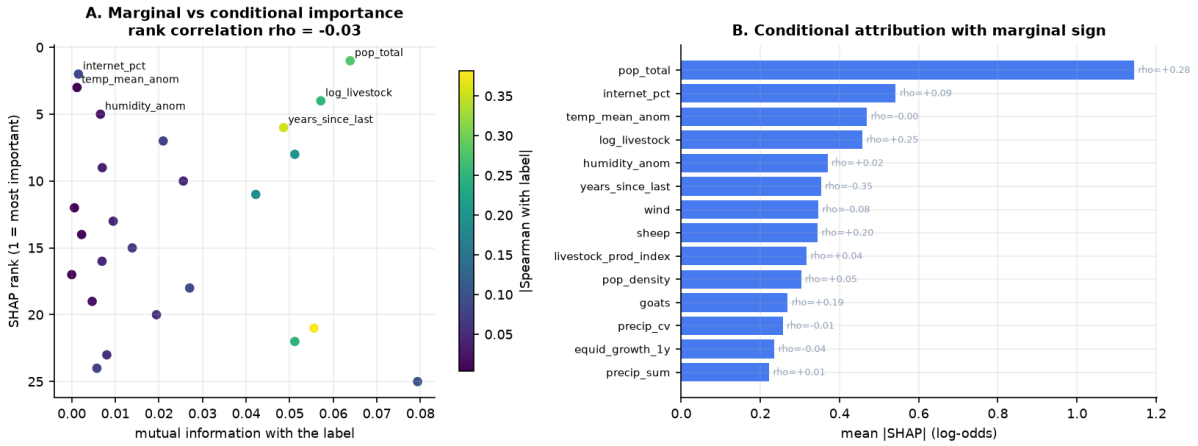


Figure 5.3: Marginal and conditional importance. Panel A plots mutual information against conditional rank, and the absence of a downward trend is the disagreement quantified in the text. Panel B gives the conditional ranking with the marginal association sign annotated.

Table 5.2: The eight highest-ranked features by mean absolute SHAP value, with marginal statistics alongside for comparison.

Rank	Feature	Mean |SHAP|	Spearman ρ	Mutual info	VIF
1	pop_total	1.144	+0.28	0.0639	4
2	internet_pct	0.542	+0.09	0.0015	5
3	temp_mean_anom	0.469	-0.00	0.0012	2
4	log_livestock	0.458	+0.25	0.0571	3
5	humidity_anom	0.370	+0.02	0.0066	2
6	years_since_last	0.353	-0.35	0.0486	2
7	wind	0.346	-0.08	0.0210	3
8	sheep	0.344	+0.20	0.0512	13

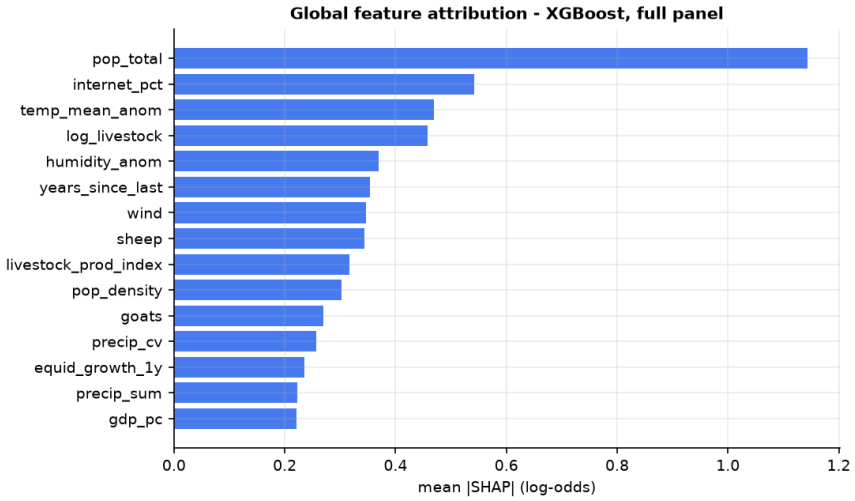


Figure 5.4: Global feature attribution for XGBoost over the full panel, ordered by mean absolute SHAP value. Population magnitude and reporting capacity dominate. Among biological drivers, recent history and climate anomalies rank highest.

3. Model Comparison

Table 5.3 reports discrimination under all three schemes. The temporal holdout is the realistic test and the leave-country-out result is the harder one.

Table 5.3: Discrimination at the country-year level. Brackets give 95 per cent bootstrap intervals on the temporal holdout.

Model	PR-AUC (time)	95% CI on PR-AUC	ROC-AUC (time)	PR-AUC (country)	PR-AUC (rep. CV)	Brier	Prec.	Rec.
Logistic regression	0.454	[0.34, 0.58]	0.812	0.223	0.414	0.0547	0.25 / 0.57	
Decision tree	0.389	[0.29, 0.50]	0.861	0.267	0.360	0.0602	0.49 / 0.53	
Random forest	0.554	[0.43, 0.66]	0.870	0.304	0.476	0.0503	0.53 / 0.50	
Extra trees	0.530	[0.42, 0.64]	0.854	0.363	0.476	0.0535	0.47 / 0.51	
XGBoost	0.485	[0.37, 0.60]	0.844	0.281	0.423	0.0536	0.61 / 0.38	
LightGBM	0.496	[0.38, 0.61]	0.841	0.339	0.397	0.0534	0.65 / 0.30	

Against a base rate of 5.59%, the strongest temporal result of 0.554 represents an approximately 10-fold lift. Every tree-based model exceeds the logistic baseline, and the single decision tree is the weakest of the six, which indicates that the relationships are interactive rather than additive.

The more important pattern is the systematic decline under leave-country-out validation. For Random Forest the score falls from 0.554 to 0.304, and the decline exceeds the width of the bootstrap interval for every family. This is the expected signature of learning country-specific structure, and it is the estimate that should be quoted for deployment.

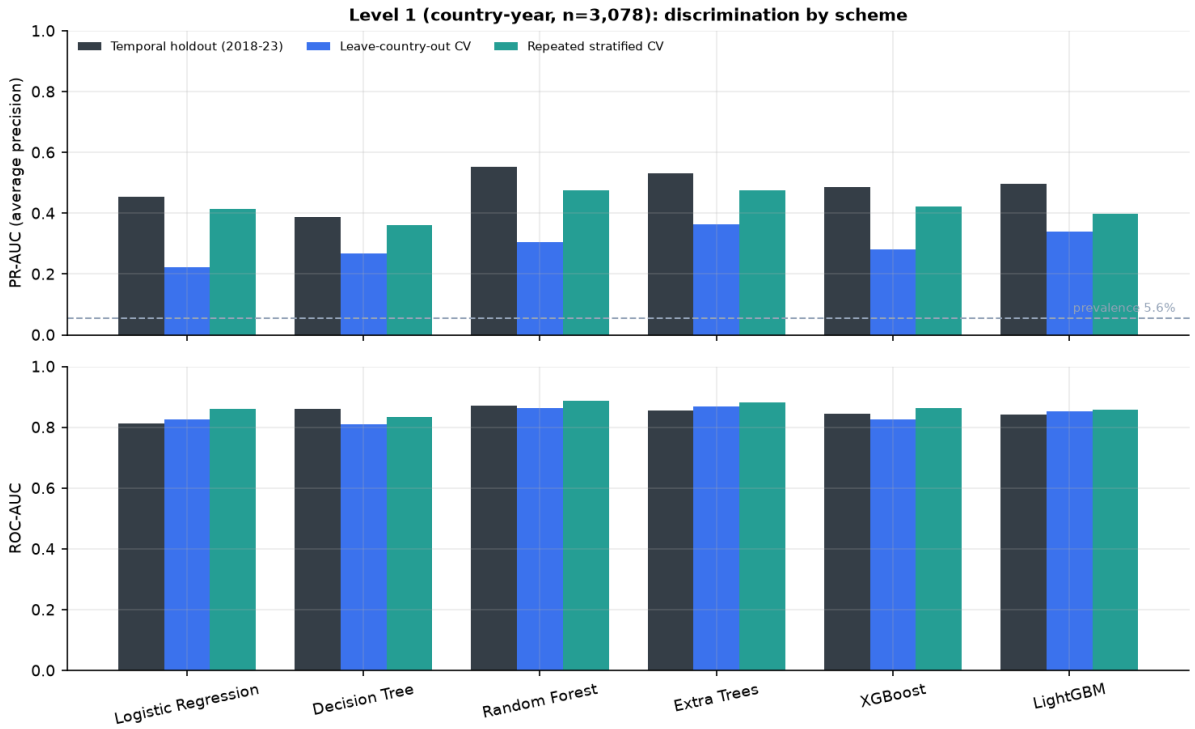


Figure 5.5: Discrimination by model and validation scheme. The gap between repeated cross-validation and leave-country-out validation is the cost of generalising to an unseen country.

Figure 5.6 shows the ROC and precision-recall curves for the temporal holdout. At this prevalence the ROC curves compress into a narrow band while the precision-recall panel separates the families.

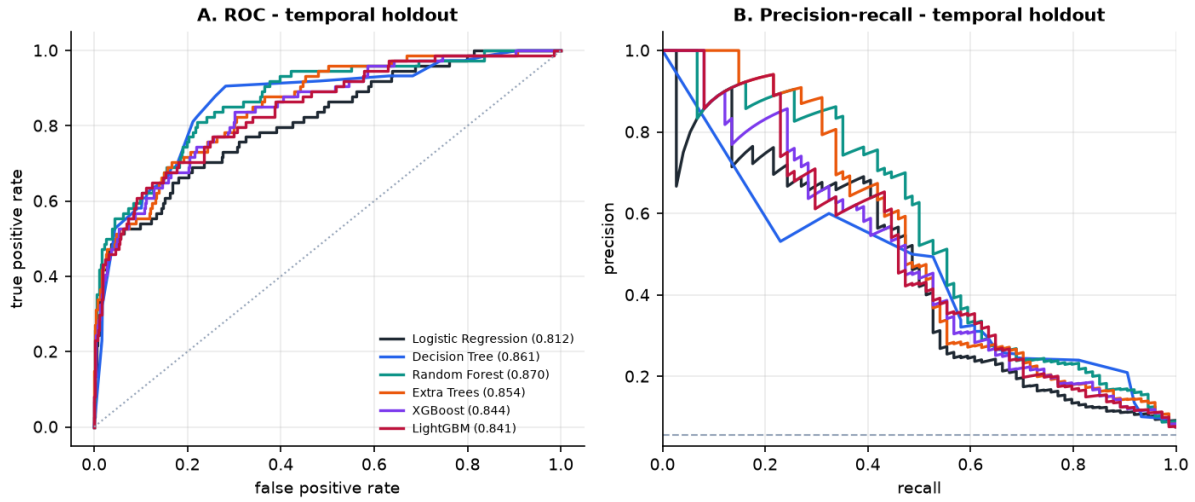


Figure 5.6: ROC curves (panel A) and precision-recall curves (panel B) on the temporal holdout.

4. Calibration

Predictions from the selected model are well calibrated, with an expected calibration error of 0.016 and a Brier score of 0.0535. A stated risk of twenty-four per cent corresponds to an observed event frequency of approximately twenty-four per cent. This property is what makes a threshold-based alert defensible, since a message quoting a probability must be numerically trustworthy.

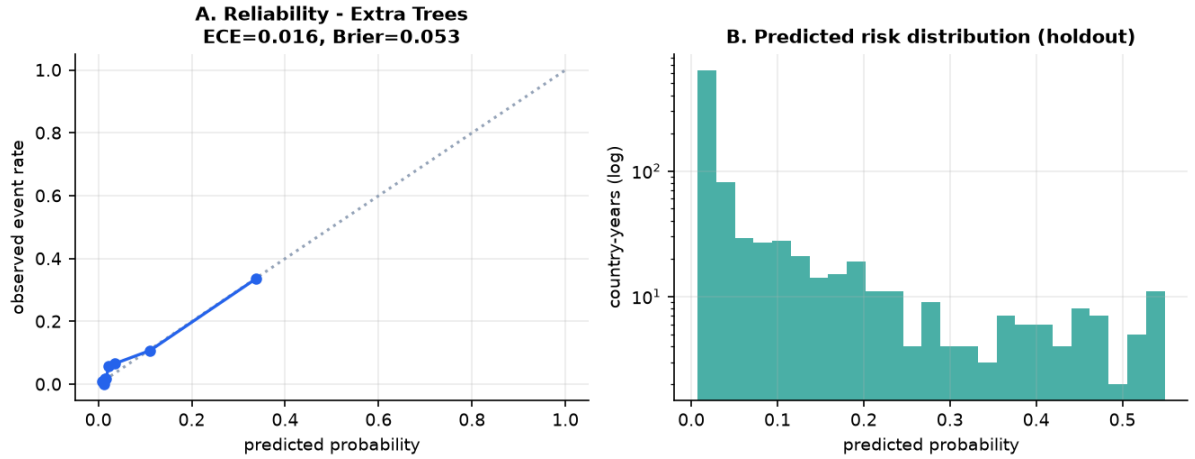


Figure 5.7: Reliability diagram for the selected model on the temporal holdout, together with the distribution of predicted risk.

5. Temporal Resolution

Aggregating observations to country and five-year period raises prevalence to 14.7% and improves every metric, as shown in Figure 5.8. The mechanism is that coarser windows match the reporting cadence of the label, and a positive anywhere within the window counts once.

6. Ablation

Feature groups were removed and each configuration retrained, with results in Table 5.4 and Figure 5.9.

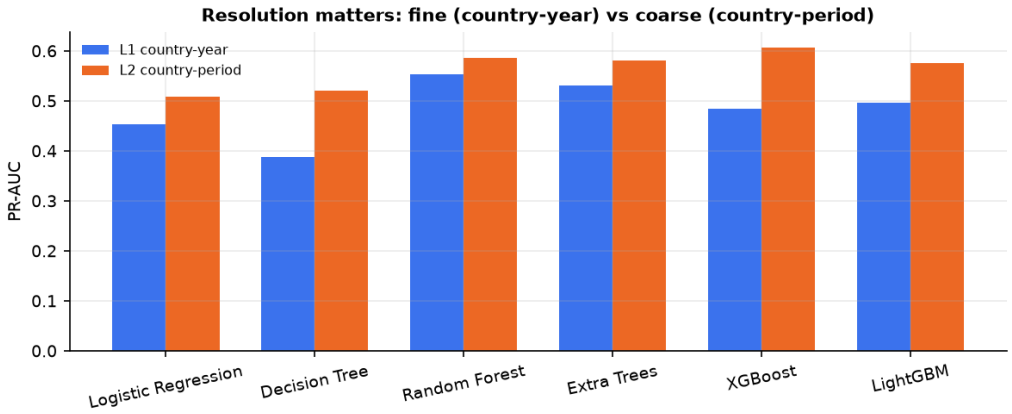


Figure 5.8: The same models evaluated at fine and coarse temporal resolution.

Table 5.4: Ablation settings, using XGBoost with fixed hyperparameters.

Feature set	Columns	PR-AUC (time)	PR-AUC (country)
History only	7	0.407	0.174
Population only	14	0.432	0.241
Climate only	10	0.395	0.078
Socio-geography only	11	0.469	0.160
History and population	21	0.451	0.321
All groups	42	0.487	0.259
All groups plus surveillance proxy	44	0.497	0.261

Two results are more informative than the headline score. First, the reduced history and population set of 21 columns reaches 0.321 under leave-country-out validation, against 0.259 for the full 42 column set. Added covariates improve in-distribution fit and reduce transfer, because they allow the model to fit country-specific peculiarities. Second, climate covariates alone reach only 0.078, marginally above the 5.59% floor, which indicates that annual country-level climatology largely encodes latitude and therefore country identity.

A separate probe adds research-output proxies as two additional columns. It moves the leave-country-out score by +0.001, which is weak evidence that the model is not purely tracking publication volume, but it is not proof.

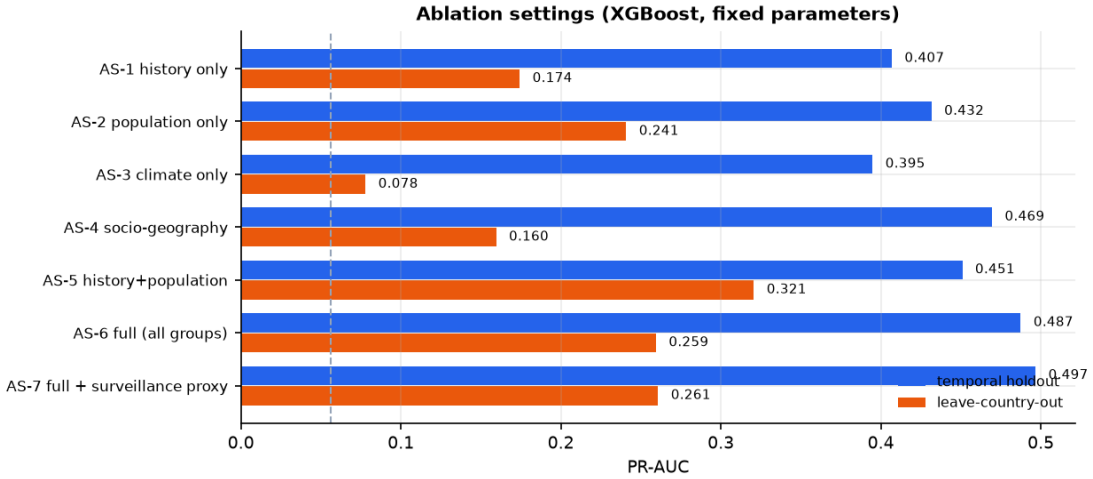


Figure 5.9: Ablation settings under both validation schemes.

7. Chained Pipeline

Table 5.5 reports the effect of appending the stage-one population forecast to the occurrence model.

Table 5.5: Effect of chaining the population forecast into the occurrence model.

Variant	PR-AUC (time)	ROC-AUC (time)	PR-AUC (country)
Plain (no chained feature)	0.530	0.854	0.363
Chained (population forecast added)	0.527	0.853	0.344

The population forecast performs well as a forecast, with a mean absolute error of 0.424 on the log scale and a Spearman correlation of 0.976. The intensity stage is weaker but usable, with a mean absolute error of 0.150 and a rank correlation of 0.465 among positive observations. The hurdle product ranks expected annual event burden with a Spearman correlation of 0.326, and its highest decile captures 70 of 106 observed events.

Appending the stage-one forecast to the occurrence model changes temporal PR-AUC by -0.003, which is inside the noise band. The explanation is structural rather than a failure of tuning. Population magnitude is already a direct feature of the occurrence model, so forecasting it and appending the forecast supplies a less accurate copy of information the model already holds. Chaining is expected to assist where the intermediate quantity cannot be observed directly, which is not the case here.

This is reported as a negative result. It is a genuine finding about chained architectures for this problem class, and it is retained rather than removed.

8. Alert Economics

The operating characteristics of the model as a warning system are shown in Figure 5.10. Alerting the six per cent of country-years with the highest predicted risk captures approximately half of all observed events, and alerting twenty-eight per cent captures eighty per cent. Uniform alerting would require full coverage for the same recall, so the model is roughly eight times more efficient than arbitrary selection at the fifty per cent operating point.

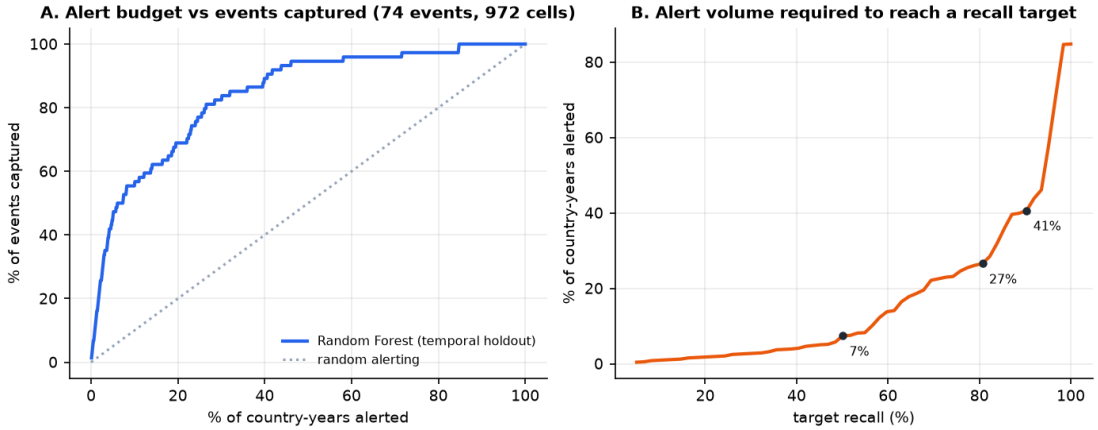


Figure 5.10: Alert economics. Panel A gives the fraction of events captured as the alert budget grows, against random alerting. Panel B gives the alert volume required to reach a target recall.

9. Subgroup Analysis

Table 5.6 reports leave-country-out performance by region. Skill is lowest in Sub-Saharan Africa and in low-income countries, which are the settings in which early warning would be most valuable. The cause is label sparsity rather than a defect in the model, since there are few recorded events from which to learn in those strata.

Table 5.6: Leave-country-out performance by World Bank region.

Region	Observations	Events	ROC-AUC	PR-AUC
East Asia & Pacific	456	48	0.865	0.396
Europe & Central Asia	912	31	0.826	0.163
Latin America & Caribbean	551	12	0.903	0.381
MENA, Afghanistan and Pakistan	380	23	0.834	0.220
Sub-Saharan Africa	665	27	0.699	0.073

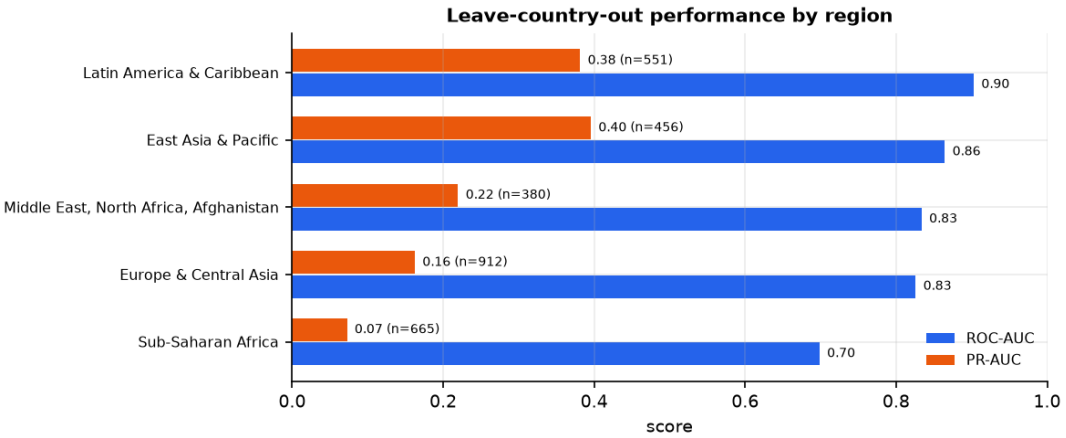


Figure 5.11: Regional performance under leave-country-out validation.

10. Discussion

The results support four conclusions.

The model has genuine predictive skill. A precision-recall AUC of 0.554 against a base rate of 5.59% is not explainable by chance, it survives a strictly forward-looking temporal split, and the probabilities are calibrated to within two percentage points.

The skill does not transfer intact. Leave-country-out performance is roughly forty per cent lower in relative terms, and this is the correct figure for planning. Any deployment to a new country should be preceded by local validation.

The principal signal is surveillance rather than biology. The evidence is consistent across three independent analyses: the highest-ranked features are population and connectivity measures, the label concentrates in countries with reference laboratories, and the best-scoring cells are India, the United States and Australia. Climate, which the original project proposal treated as a primary input, transfers poorly at this resolution.

Simpler models transfer better. The reduced feature set outperforms the full set across countries, and the linear baseline degrades least in absolute terms. This is a practical argument for parsimony when positive examples are limited.

Chapter 6

Conclusion and Future Work

1. Conclusion

This project set out to determine whether population-level glanders risk could be predicted from publicly available data. Four sources that had not previously been applied to the disease were assembled into a country-year panel covering 162 equine-holding countries, and six model families were trained and evaluated under three validation schemes, together with a three-stage chained pipeline.

The central finding is that prediction is possible, and that the achievable performance is moderate. On a strictly forward-looking temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 5.59%, an approximately 10-fold lift, with an expected calibration error of 0.016. Used as a warning system, it captures half of all observed events while alerting six per cent of country-years.

The second finding is that the skill is fragile across geography. Leave-country-out performance is approximately forty per cent lower in relative terms, and the decline is concentrated in the regions where early warning would be most useful.

The third finding concerns what the model is learning. Three independent analyses converge on the same conclusion. The highest-ranked features are population size and reporting capacity rather than biological drivers. The label concentrates in countries that operate reference laboratories. And the highest-scoring cells are the countries with the strongest veterinary publishing record. The model predicts where glanders will be documented, which is useful for directing surveillance, and it is not a map of where the organism is present.

The fourth finding is negative. The chained design specified in the project proposal does not improve discrimination, because the intermediate quantity is already available as a direct feature. This is reported rather than omitted, since it is a reproducible result about chained architectures for this problem class.

2. Future Work

Four directions follow directly from the limitations identified in this report.

Modelling reporting effort explicitly. The most important extension is to treat surveillance capacity as a modelled quantity rather than an incidental covariate. This could be done by using publication volume as a separate offset, or by fitting a two-part model in which detection and occurrence are estimated jointly. Without this, gains in predictive score are likely to reflect reporting practice rather than disease.

Validating the label extractor. The 263 extracted events have not been checked against a hand-annotated reference set. Annotating a stratified sample of several hundred abstracts would allow precision and recall of the extractor itself to be reported, and would bound the label noise that currently

propagates into every downstream estimate.

District-level modelling for India. The original proposal envisaged district-month resolution for India, where the disease is concentrated in a small number of states. Public district-level covariates for equine population and climate would need to be assembled, but the resolution would match the scale at which control measures are applied.

Operational calibration under drift. A deployed model requires periodic recalibration as reporting practice changes. The calibration framework used here is a single point estimate; a production system would need drift monitoring and a defined retraining trigger.

Two smaller extensions are worth noting. Incorporating WOAH WAHIS official outbreak records as a second label source would allow the literature-derived label to be validated where the two overlap. And a cost-sensitive threshold, in which the relative cost of a missed outbreak and a false alert is stated explicitly, would replace the F1-optimal threshold used here with one matched to the programme's actual priorities.

Bibliography

[1] Khan I, Wieler LH, Melzer F, et al. Glanders in animals: a review on epidemiology, clinical presentation, diagnosis and countermeasures. *Transboundary and Emerging Diseases*. 2013. doi:10.1111/j.1865-1682.2012.01342.x

[2] Kanani Y, Oryan A, Derakhshandeh A. Glanders: past, present and future of a neglected zoonosis as a threat in veterinary and human health. *Veterinary Medicine and Science*. 2026. doi:10.1002/vms3.70950

[3] Purnama MTE, Firdausy LW, Fikri F, et al. Seroprevalence and associated epidemiological drivers of glanders in horses: a systematic review and meta-analysis. *Research in Veterinary Science*. 2026;106416. doi:10.1016/j.rvsc.2026.106416

[4] Mota RA, Pinheiro Júnior JW. Current status of glanders in Brazil: recent advances and challenges. *Brazilian Journal of Microbiology*. 2022. doi:10.1007/s42770-022-00814-1

[5] Singha H, Shanmugasundaram K, Jhandai P, et al. Surveillance of equine glanders in India (2019–2023): prevalence estimate, risk analysis and future strategy. *Microbial Pathogenesis*. 2026;217:108597. doi:10.1016/j.micpath.2026.108597

[6] Singha H, Shanmugasundaram K, Tripathi BN, et al. Serological surveillance and clinical investigation of glanders among indigenous equines in India from 2015 to 2018. *Transboundary and Emerging Diseases*. 2020. doi:10.1111/tbed.13475

[7] Malik P, Singha H, Khurana SK, et al. Emergence and re-emergence of glanders in India: a description of outbreaks from 2006 to 2011. *Veterinaria Italiana*. 2012. PMID:22718333

[8] Fonseca-Rodríguez O, Pinheiro Júnior JW, Mota RA. Spatiotemporal analysis of glanders in Brazil. *Journal of Equine Veterinary Science*. 2019. doi:10.1016/j.jevs.2019.03.216

[9] Hollings T, Robinson A, van Andel M, et al. Species distribution models: a comparison of statistical approaches for livestock and disease epidemics. *PLOS ONE*. 2017;12(8):e0183626. doi:10.1371/journal.pone.0183626

[10] Alshatti M, Al-Hemoud A, Othman A, et al. A risk prediction model for highly pathogenic avian influenza outbreaks in Kuwait. *Frontiers in Veterinary Science*. 2026. doi:10.3389/fvets.2026.1843962

[11] Kane MJ, Price N, Scotch M, Rabinowitz P. Comparison of ARIMA and Random Forest time series models for prediction of avian influenza H5N1 outbreaks. *BMC Bioinformatics*. 2014;15:276. doi:10.1186/1471-2105-15-276

[12] Walsh DP, Ma TF, Ip HS, Zhu J. Artificial intelligence and avian influenza: using machine learning to enhance active surveillance for avian influenza viruses. *Transboundary and Emerging Diseases*. 2019. doi:10.1111/tbed.13318

[13] Yoo DS, Song YH, Choi DW, et al. Machine learning-driven dynamic risk prediction for highly pathogenic avian influenza at poultry farms in the Republic of Korea. *Transboundary and Emerging Diseases*. 2021. doi:10.1111/tbed.14419

[14] Moawad AA, Silge A, Bocklitz T, Fischer K, Rösch P, Roesler U, Elschner MC, Popp J, Neubauer H. A machine learning-based Raman spectroscopic assay for the identification of *Burkholderia mallei* and related species. *Molecules*. 2019;24(24):4516. doi:10.3390/molecules24244516

[15] Stöckel S, Meisel S, Elschner M, Melzer F, Rösch P, Popp J. Raman spectroscopic detection and identification of *Burkholderia mallei* and *Burkholderia pseudomallei*. *Analytical and Bioanalytical Chemistry*. 2015;407(3):833–843. doi:10.1007/s00216-014-7906-5

[16] Collier N. Uncovering text mining: a survey of current work on web-based epidemic intelligence. *Global Public Health*. 2012. doi:10.1080/17441692.2012.699975

[17] Lejeune G, Brixtel R, Doucet A, et al. Multilingual event extraction for epidemic detection. *Artificial Intelligence in Medicine*. 2015. doi:10.1016/j.artmed.2015.06.005

[18] Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016:785–794. doi:10.1145/2939672.2939785

[19] Ke G, Meng Q, Finley T, et al. LightGBM: a highly efficient gradient boosting decision tree. In: *Advances in Neural Information Processing Systems 30 (NIPS)*.

2017:3146–3154.

[20] Breiman L. Random forests. *Machine Learning*. 2001. doi:10.1023/A:1010933404324

[21] Lundberg SM, Lee SI. A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems 30 (NIPS)*. 2017:4765–4774. arXiv:1705.07874

[22] Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*. 2015;10(3):e0118432. doi:10.1371/journal.pone.0118432

[23] Davis J, Goadrich M. The relationship between precision-recall and ROC curves. In: *Proceedings of the 23rd International Conference on Machine Learning*. 2006:233–240. doi:10.1145/1143844.1143874

[24] Roberts DR, Bahn V, Ciuti S, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*. 2017. doi:10.1111/ecog.02881

[25] Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. In: *Proceedings of the 22nd International Conference on Machine Learning*. 2005:625–632. doi:10.1145/1102351.1102430

[26] Huang Y, Li W, Macheret F, et al. A tutorial on calibration measurements and calibration models for clinical prediction models. *Journal of the American Medical Informatics Association*. 2020;27(4):621–633. doi:10.1093/jamia/ocz228

[27] WOAH. World Animal Health Information System (WAHIS). https://wahis.woah.org

[28] Department of Animal Husbandry and Dairying, Government of India. *National Action Plan for Control and Eradication of Glanders in India* (Revised 2025). New Delhi: DAHD; 2025.

[29] FAO. FAOSTAT: crops and livestock products. https://www.fao.org/faostat

[30] NASA Langley Research Center. POWER agroclimatology API. https://power.larc.nasa.gov

[31] World Bank. World Development Indicators. https://databank.worldbank.org

[32] Europe PMC literature search API. https://europepmc.org